

EXPERIMENT - 13

Title

Write a Program to Create a Simple Calculator Application using ReactJS

Aim

To develop a **Simple Calculator Application** using ReactJS that performs basic arithmetic operations such as:

- Addition
 - Subtraction
 - Multiplication
 - Division
-

Prerequisites

- Basic knowledge of JavaScript (ES6)
 - Understanding of ReactJS components
 - Node.js installed
 - npm installed
-

Software Requirements

- OS: Windows / Linux / macOS
- Node.js (LTS)

- ReactJS
 - VS Code
 - Browser (Chrome)
-

Theory

React is a JavaScript library used to build user interfaces using components.

Key Concepts Used:

- **Component** → Reusable UI block
 - **useState Hook** → Manages state
 - **Event Handling** → onClick()
 - **JSX** → HTML inside JavaScript
-

Project Setup

Step 1: Create React App

```
Shell
npm create vite@latest
cd calculator-app
npm start
```

```
> npx
> create-vite
```

```

◇ Project name:
  calculator-app

◇ Select a framework:
  React

◇ Select a variant:
  JavaScript

◇ Use Vite 8 beta (Experimental)?:
  No

◇ Install with npm and start now?
  Yes

```

Source Code

Replace the content of `src/App.js` with the following:

JavaScript

```
import React, { useState } from "react";
import "./App.css";

function App() {
  const [input, setInput] = useState("");
}
```

```

const handleClick = (value) => {
  if (value === "=") {
    try {
      // ⚠️ Avoid eval in production applications
      setInput(eval(input).toString());
    } catch (error) {
      setInput("Error");
    }
  } else if (value === "C") {
    setInput("");
  } else {
    setInput(input + value);
  }
};

return (
  <div className="App">
    <h1>React Calculator</h1>

    <div className="calculator">
      <div className="display">{input || "0"}</div>

      <div className="buttons">
        {[ "7", "8", "9", "/",
          "4", "5", "6", "*",
          "1", "2", "3", "-",
          "0", ".", "C", "+" ].map((btn) => (
            <button key={btn} onClick={() =>
handleButtonClick(btn)}>
              {btn}
            </button>
          )))}
      </div>
    </div>
  </div>
);

```

```
        <button
          className="equals"
          onClick={() => handleButtonClick("=")}
        >
          =
        </button>
      </div>
    </div>
  </div>
);
}

export default App;
```

App.css

Replace `src/App.css` with:

```
CSS

.App {
  text-align: center;
  font-family: Arial, sans-serif;
}

.calculator {
  display: inline-block;
  background: #f0f0f0;
  padding: 20px;
  border-radius: 10px;
```

```
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);  
}
```

```
.display {  
    width: 200px;  
    height: 40px;  
    margin-bottom: 10px;  
    background: #fff;  
    text-align: right;  
    padding: 10px;  
    border: 1px solid #ccc;  
    border-radius: 5px;  
    font-size: 18px;  
    overflow-x: auto;  
}
```

```
.buttons {  
    display: grid;  
    grid-template-columns: repeat(4, 50px);  
    gap: 10px;  
}
```

```
button {  
    width: 50px;  
    height: 50px;  
    font-size: 18px;  
    background: #007bff;  
    color: white;  
    border: none;  
    border-radius: 5px;  
    cursor: pointer;  
    transition: background 0.3s;  
}
```

```
button:hover {  
  background: #0056b3;  
}  
  
.equals {  
  grid-column: span 4;  
  background: #28a745;  
}  
  
.equals:hover {  
  background: #1e7e34;  
}
```

Execution Steps

1. Run project:

```
Shell  
npm start
```

2.
Open browser:

```
None  
http://localhost:3000
```

Functionality Demonstrated

Operation	Button
Addition	Add
Subtraction	Subtract
Multiplication	Multiply
Division	Divide

Sample Output

Input:

```
None
10 and 5
```

Output:

```
None
Result: 15 (Addition)
Result: 5 (Subtraction)
Result: 50 (Multiplication)
Result: 2 (Division)
```

Expected Result

The ReactJS calculator successfully performs arithmetic operations using state management and event handling.

Result

Thus, a Simple Calculator Application was successfully developed using ReactJS.

Viva Voce Questions

1. What is React?
 2. What is JSX?
 3. What is useState?
 4. What is component in React?
 5. What is event handling in React?
 6. Why do we use parseFloat?
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Conclusion

This experiment demonstrates ReactJS fundamentals including components, state management, and event handling by implementing a functional calculator application.